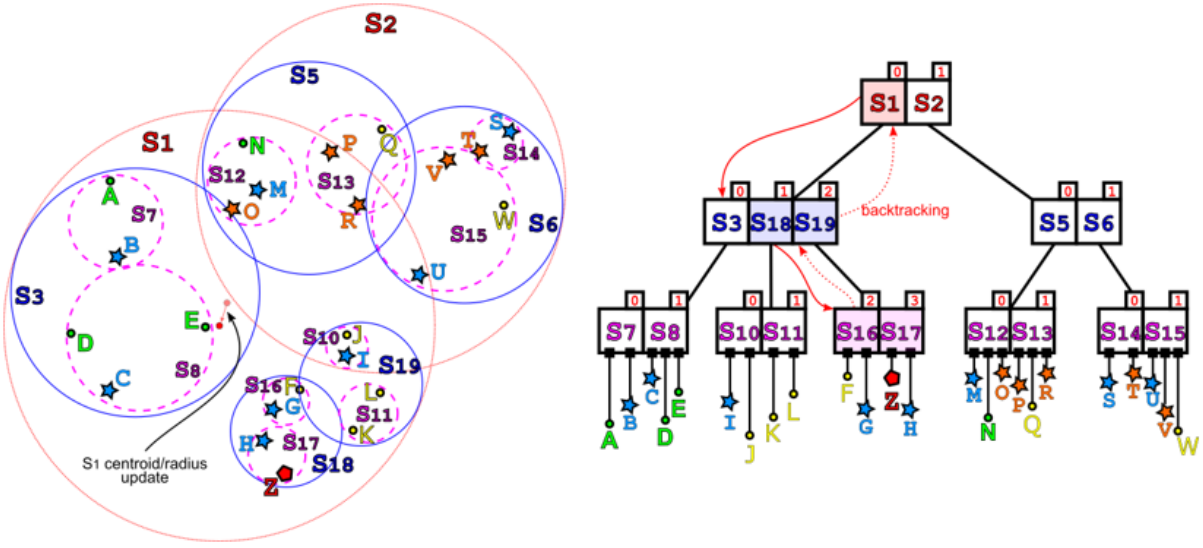


Figure 10.15 Backtracking in method `insert` after splitting a leaf. Continuing from figure 10.14, after we split leaf S_9 into nodes S_{16} and S_{17}, we backtrack to S_9's parents, S_4, and add these two new leaves to it: as shown at the end of figure 10.14, S_4 now has 4 children, one too many: we need to split it as well. Here we show the result of splitting S_4 into two new nodes, S_{18} and S_{19}, that will be added to S_4's parent, S_1. To which, in turn, we will backtrack: since it now has only 3 children (and `M==3`) we just recompute centroid and radius for S_1's bounding envelope, and we can stop backtracking.

In listing 10.6 we used several helper functions[9] to perform insertion; however, there is still one case that is not handled: what happens when we get to the root, and we need to split it?

The reason for not handling this case as part of the method in listing 10.6 is that we would need to update the root of the